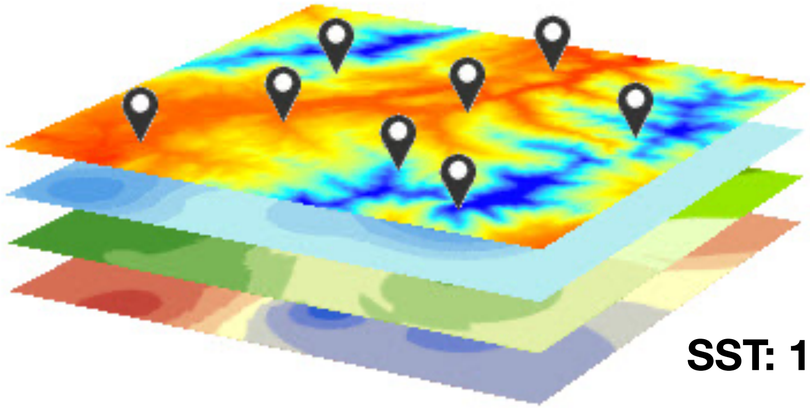
Session 2: Spatial foundations, data acquisition and quality

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Biological Data for SDMs



Occurrence records for modeling

Occurrence records allow building SDMs and answer *why* a species occurs in a given area.

Essential components: Data must **reference a particular taxon**, at a **particular place** (geographic coordinates), and (ideally) at a **particular time**.

Metadata can describe the source of record (sampling method), means of identification (e.g., visual census), and spatial precision of record (maximum error).



Occurrence data for SDM: one species example

Data

Lon	Lat	Date	Depth	Species
0,61287	39,58955	10/08/12	22	Caretta caretta
0,83999	39,77417	03/03/12	12	Caretta caretta
3,18192	42,40479	14/08/11	1	Caretta caretta
-4,86395	55,24295	01/01/04	0	Caretta caretta
-5,89297	57,06977	03/03/12	0	Caretta caretta
34,9082	32,55822	22/12/02	23	Caretta caretta
0,61287	39,58955	14/08/11	12	Caretta caretta
0,83999	39,77417	03/03/12	2	Caretta caretta
3,18192	42,40479	22/12/02	5	Caretta caretta
-4,86395	55,24295	03/03/12	7	Caretta caretta
-5,89297	57,06977	01/01/04	3	Caretta caretta

Metadata

Common name	Behaviour	Observer
Loggerhead sea turtle	Swimming	John Doe
Loggerhead sea turtle	Swimming	John Doe
Loggerhead sea turtle	Eating	Unknown
Loggerhead sea turtle	Swimming	Unknown
Loggerhead sea turtle	Sleeping	John Doe
Loggerhead sea turtle	Sleeping	John Doe
Loggerhead sea turtle	Swimming	John Doe
Loggerhead sea turtle	Sleeping	John Doe
Loggerhead sea turtle	Eating	Unknown
Loggerhead sea turtle	Swimming	Unknown
Loggerhead sea turtle	Eating	Unknown

Where to Source Biological Data

Global Biodiversity information centres;

Peer-reviewed literature;

Reference collections (e.g. museums and herbaria);

Citizen science initiatives.

Occurrence records: Global Biodiversity information centers

(i.e., databases linked via a common access portal).

Global Biodiversity Information Facility (GBIF; https://www.gbif.org/):

Occurrence records: over 3 billion

Ocean Biodiversity Information System (OBIS; https://obis.org/):

Occurrence records: over 200 million

National Biodiversity Network (NBN; UK; https://nbnatlas.org/):

Occurrence records: over 320 million

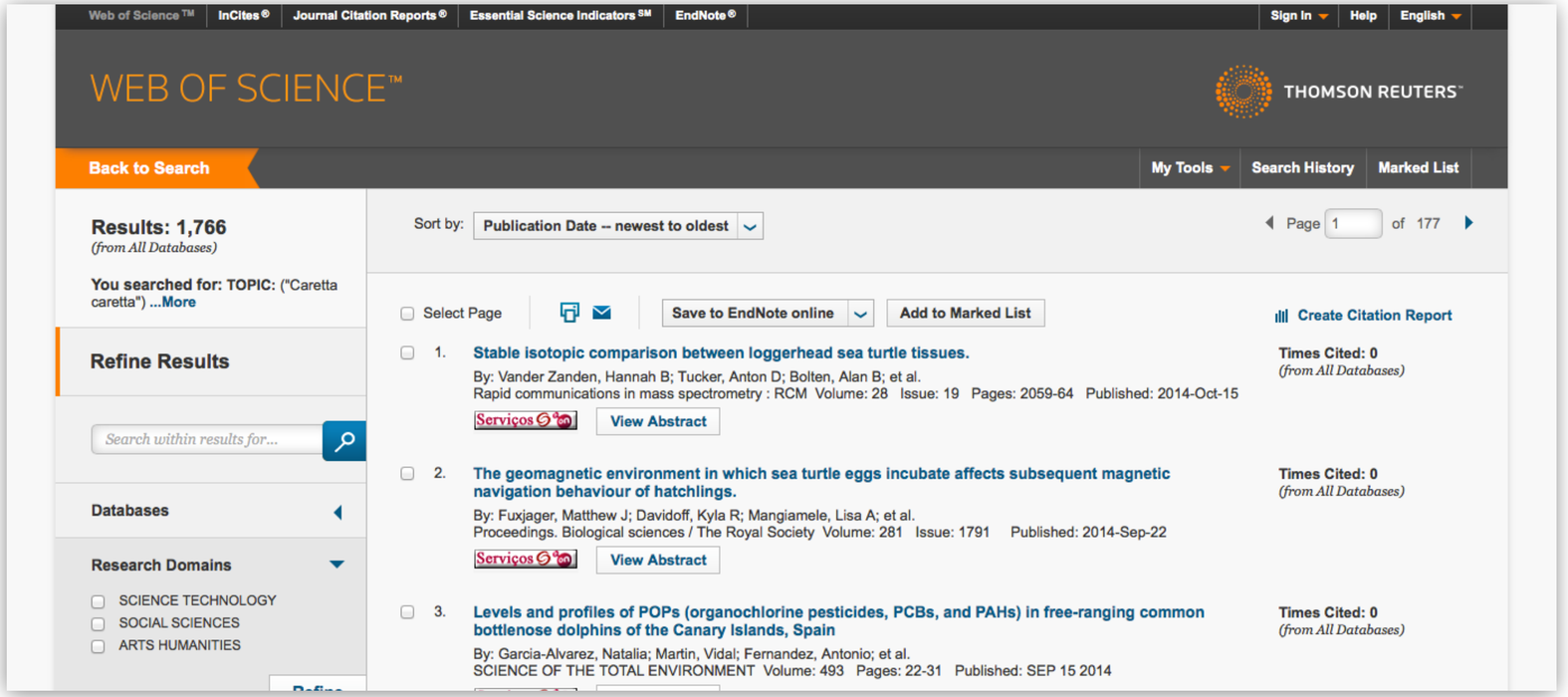
Atlas of Living Australia (ALA; https://www.ala.org.au/):

Occurrence records: over 130 million

Occurrence records: Literature

Literature is useful to compile biodiversity records.

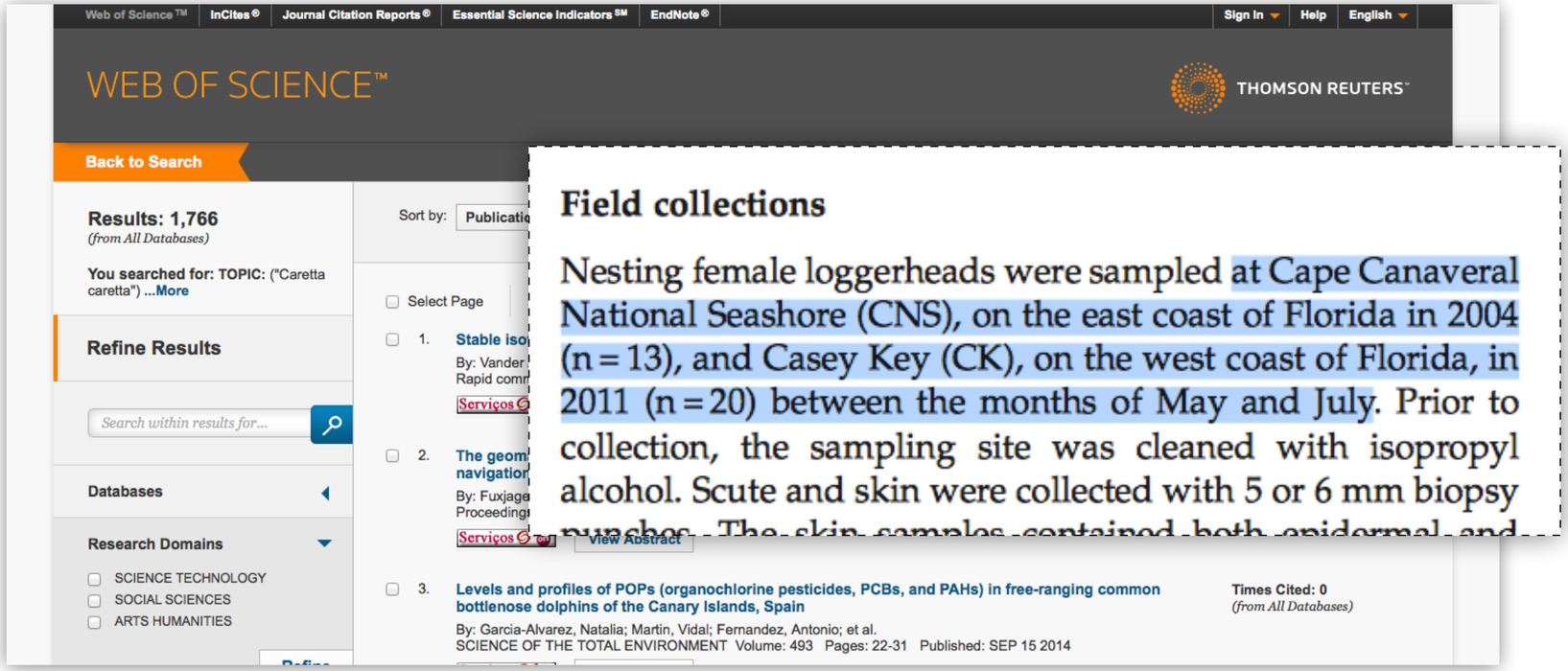
Peer-reviewed publications are preferred, even when out of the scope of our work.



Occurrence records: Literature

Literature is useful to compile biodiversity records.

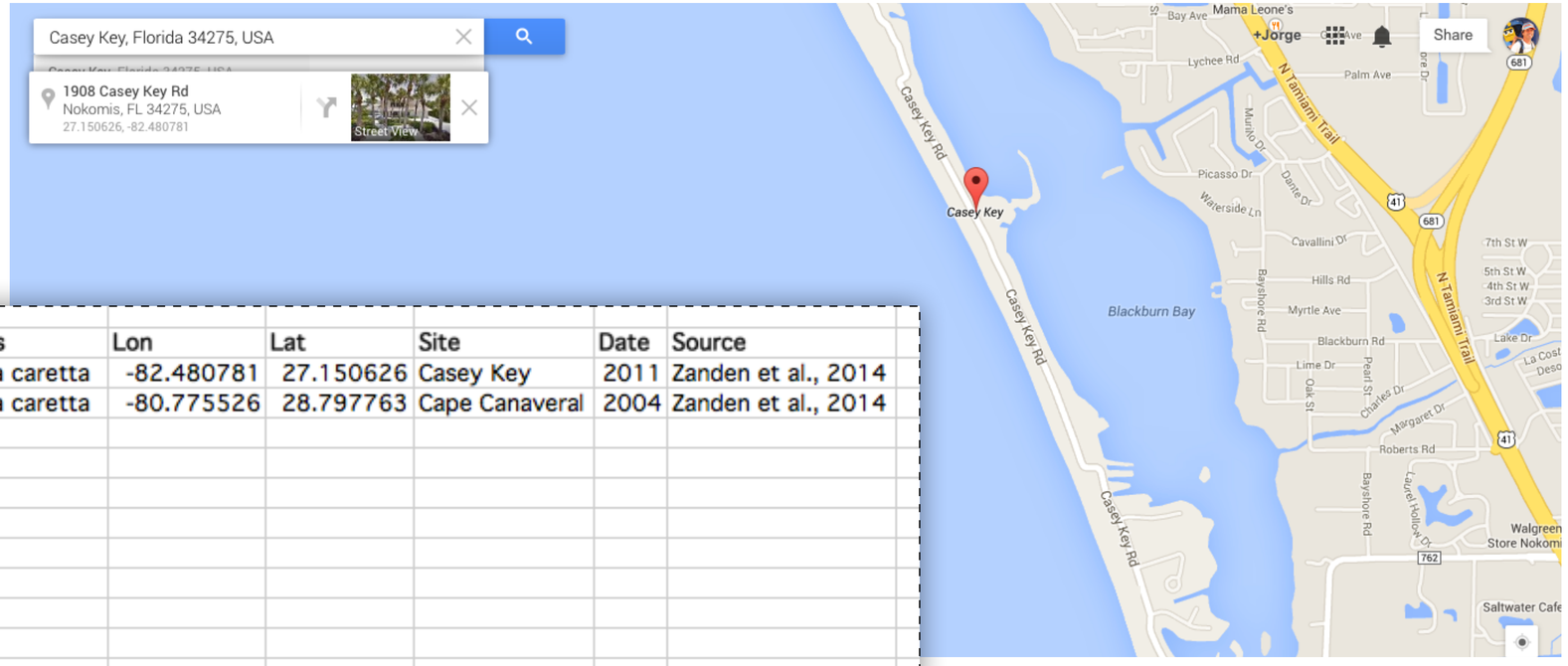
Peer-reviewed publications are preferred, even when out of the scope of our work.



Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.



Species	Lon	Lat	Site	Date	Source
Caretta caretta	-82.480781	27.150626	Casey Key	2011	Zanden et al., 2014
Caretta caretta	-80.775526	28.797763	Cape Canaveral	2004	Zanden et al., 2014

Occurrence records: Biological reference collections

Preserved specimens / **photographs** maintained in **museums**, **universities**, **botanic gardens**, and similar **institutions**.

A major tools for basic investigation and assessment of biodiversity.

~2.5 billion specimens in preserved collections.

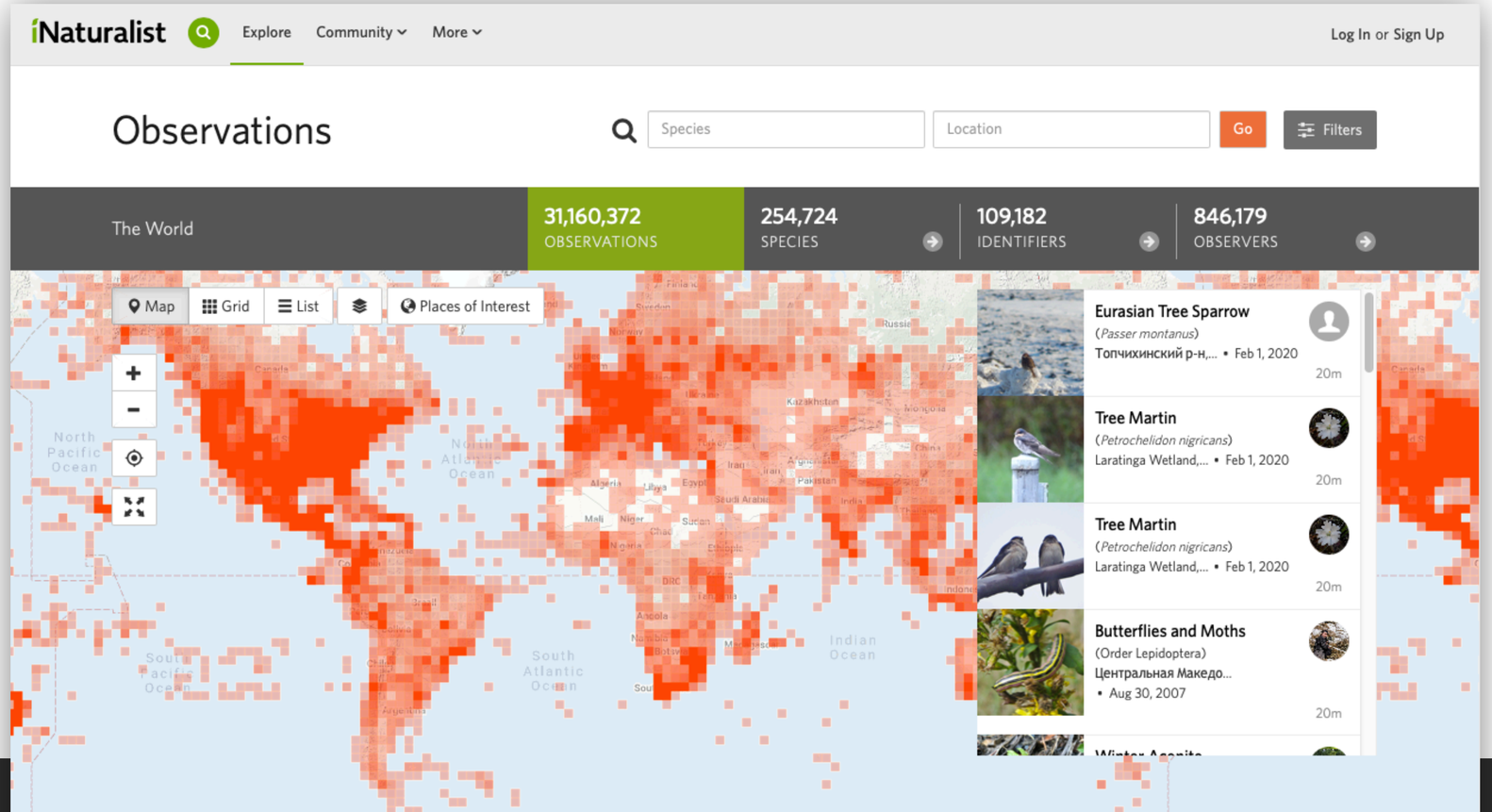
Macroalgae Herbarium Portal (www.macroalgae.org):

Occurrence records: >1 million records.

Occurrence records: Citizens can help

Sharing biodiversity records / pictures with scientists.

iNaturalist (http://www.inaturalist.com): > 300 million records.



2 min. PITCH

Papers: P59, P60, P66

10 minutes

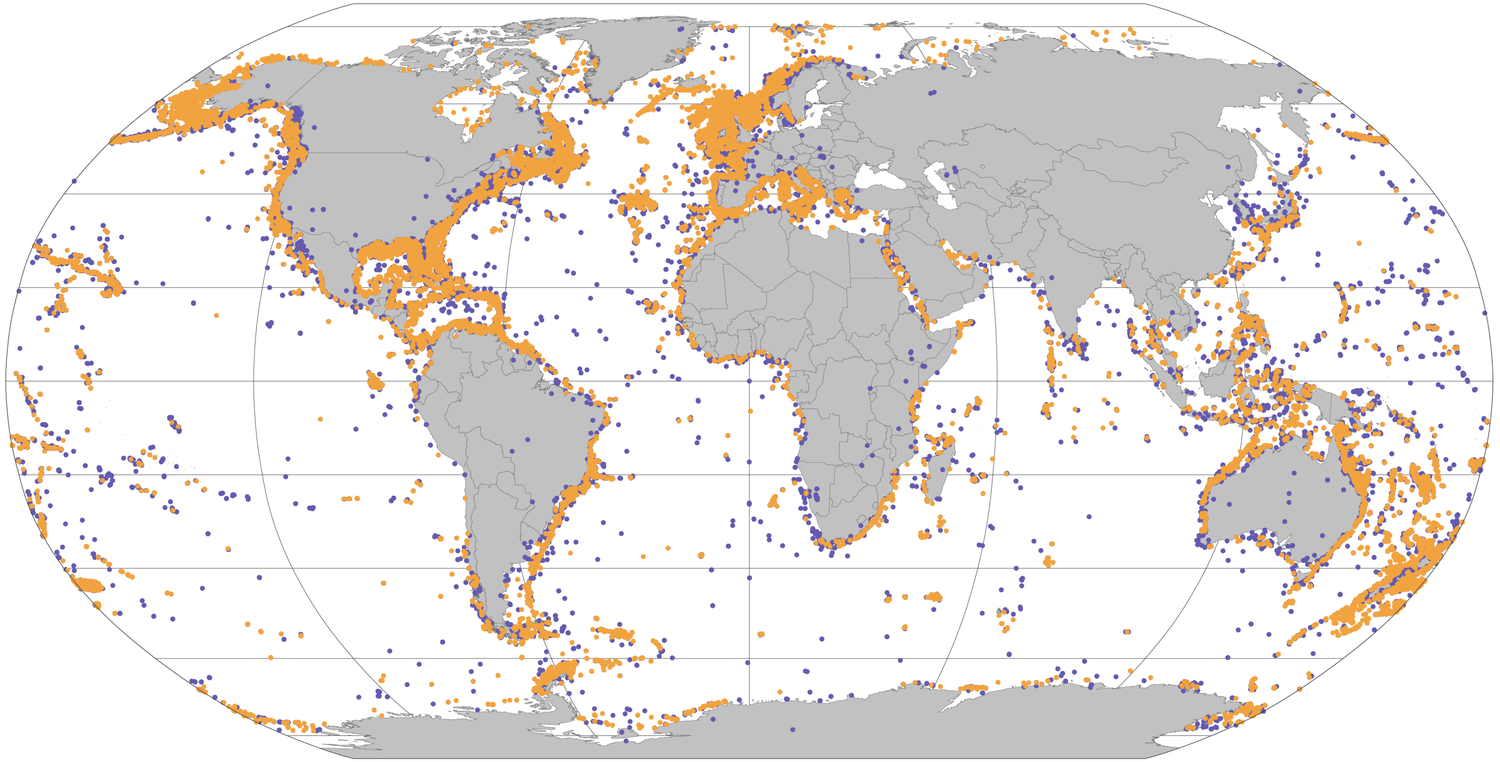
Read the abstract and the introduction and interpret the main tables.

Question. What problem did the authors tackle? (one sentence)

Approach. What did they do, and with which data? (one sentence)

Key finding. The one result to remember, ideally with the paper's main figure.

Why it matters today?



4 million records of ~1,500 species of kelp, seagrass and cold-water corals.

Data gathered from **literature** and major **biodiversity information facilities**.

Quality control of records based on published and expert knowledge to **eliminate entries outside the known vertical and geographic distribution of species.**

Assis et al., 2020